Mohamed Zayed Houachmi

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Education

University of Ottawa — Ottawa, ON
B.A.Sc. Mechanical Engineering (Co-op) — *Sept. 2025 – June 2029*

Experience

Renewable Energy Intern — June – Aug. 2026
Greenard Consulting — *Casablanca, Morocco*

- Ran a full wind-resource assessment in **WAsP**: resource grids, turbine siting, AEP and wake-loss evaluation
- Built an **8-module Python/Streamlit** tool implementing Jensen and Ainslie wake models via Runge-Kutta
- Validated against WAsP to a 4.90 m/s mean wind speed; documented methodology and limitations
- Quantified a **43% wake loss** (16,186 to 9,264 MWh/year) by modelling a 6-turbine, 1 MW site at 5D spacing

Formula SAE — Aerodynamics Team — Sept. 2025 – Present
uOttawa Formula One Student Group — *Ottawa, ON*

- Designed front and rear wing assemblies in SolidWorks within FSAE rules-box and packaging constraints
- Ran full-car CFD predicting **320 N downforce** at 65 km/h at a coefficient of lift of –1.91
- Iterated wings, undertray, diffuser and sidepods in **ANSYS CFD**; front wing reached Cl –0.800, Cd 0.109
- Validated CFD against suspension-travel data: measured downforce within **10%** of prediction at 65 km/h
- Manufactured aerodynamic surfaces by composite layup: carbon fibre mould prep, layup, curing and assembly

Founder — Aug. 2026 – Present
Tutoring Ottawa — *Ottawa, ON*

- Founded and operate a bilingual tutoring business for Grades 7–12; manage client acquisition directly
- Built the booking platform in **Next.js/TypeScript**: server-validated API, Supabase/Postgres with row-level security
- Hardened intake with rate limiting, honeypot filtering, input sanitization and consent capture

Client Services Representative — June 2025 – Present
Norton Rose Fulbright — *Ottawa, ON*

- Coordinate logistics and client communication for corporate meetings under confidentiality requirements
- Manage competing requests independently in a deadline-driven, accuracy-critical environment

Junior STEM Instructor — May 2026 – Present
Young Engineers — *Ottawa, ON*

- Teach mechanics, structures and problem-solving to young learners through hands-on build projects

Projects

Blood-Sample Swab Drying System | *Team Lead, Ottawa Police Service client* — Oct. – Dec. 2025

- Led a student team through design, fabrication, assembly and testing of a drying enclosure
- Delivered a functional prototype meeting the client's primary objective
- Created SolidWorks parts and assemblies; fabricated components with 3D printing and laser cutting
- Integrated Arduino temperature and humidity sensors; analyzed drying-test data and reported findings

2D Platformer Game | *C#, .NET WinForms* — 2024

- Implemented velocity-based gravity and per-axis collision resolution across five levels

E-Commerce Web Application | *JavaScript, HTML/CSS, Supabase* — 2025

- Built a six-category storefront with product pages, cart and accounts on Postgres, no frameworks

Leadership

Founder & President, STEM Club — Feb. 2025 – June 2025
De La Salle High School — *Ottawa, ON*

- Founded a student organization and grew it past **50 members** in one semester with faculty backing
- Presented to roughly **1,000** students and staff; ran a peer-tutoring program for 10–30 students

Technical Skills

Engineering: SolidWorks, ANSYS CFD, WAsP, numerical modelling and validation, Arduino, composite layup
Languages: Python (NumPy, Streamlit), C, C++, C#, JavaScript, HTML/CSS, Visual Basic
Web: Next.js, TypeScript, React, Supabase (Postgres), Vercel
Software: Microsoft Excel, Word, PowerPoint
Spoken: English (fluent), French (fluent, DELF certified), Arabic (native)